LAB 7: Fuzzing for Input Validation Bugs

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Assigned Repository: L6_L7_60_marathon_registration **Module Under Test:** marathon_registration_parser.py

1. Objective

To explore fuzz testing using the Hypothesis property-based testing framework, identify input validation bugs and unhandled crashes in the assigned Python module (marathon_registration_parser.py), and apply defensive programming techniques to ensure the module handles arbitrary and malformed inputs gracefully.

2. Test Execution and Screenshots

Screenshot 1 (SS1): Running Buggy Application with Valid Inputs

Command: python marathon_registration_parser.py. The baseline buggy script executes successfully with clean, well-formed user input.

```
PS D:\Software Engineering\SE-Labs-PES1UG24CS928\Lab-7\submissions\L6_L7_60_marathon_registration\LAB7> python maratho
n_registration_parser.py
==== Running format_bib_name ====
Enter the name to print on the bib: Ritu Ravish
Bib name: RITU RAVISH

==== Running parse_finish_time ====
Enter a finish time as H:MM:SS (e.g. 3:45:10): 3:45:10
Finish time in seconds: 13510

==== Running average_split ====
Enter your 5 km split times in minutes separated by spaces (e.g. 25 26 24): 25 26 24
Average split (minutes): 25.0
```

Screenshot 2 (SS2): Fuzz Testing Buggy Application with Hypothesis

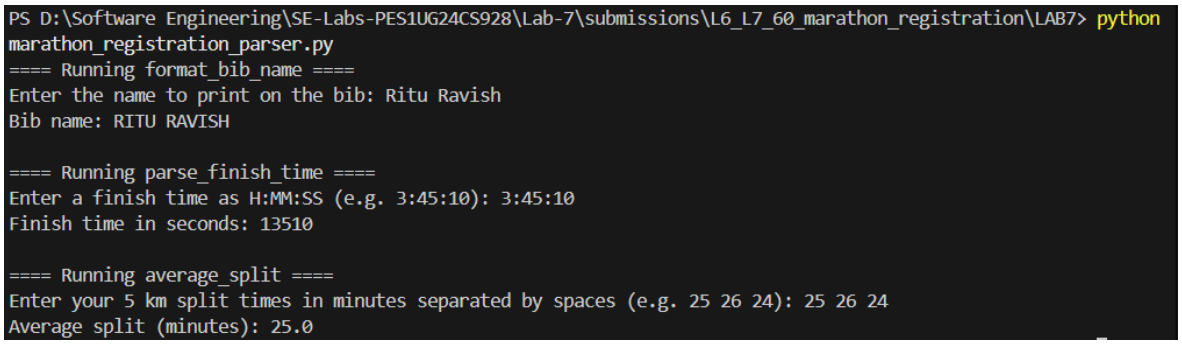
Command: python -m pytest test_marathon_registration_parser.py. Hypothesis uncovers edge cases (None, empty strings, missing delimiters, non-numeric values), causing all three functions to crash with unhandled exceptions.

```
============================================= warnings summary =============================================
C:\Users\Ritu Ravish\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\si
te-packages\anyio\pytest_plugin.py:14
  C:\Users\Ritu Ravish\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\
site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================== short test summary info ==========================================
FAILED test_marathon_registration_parser.py::test_format_bib_name_no_crash - AssertionError: format_bib_name crashed with: 'NoneType'
 object has no attribute 'strip'
FAILED test_marathon_registration_parser.py::test_parse_finish_time_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures.
(2 sub-exceptions)
FAILED test_marathon_registration_parser.py::test_average_split_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 s
ub-exceptions)
======================================== 3 failed, 1 warning in 5.92s ========================================
PS D:\Software Engineering\SE-Labs-PES1UG24CS928\Lab-7\submissions\L6_L7_60_marathon_registration\LAB7>
```

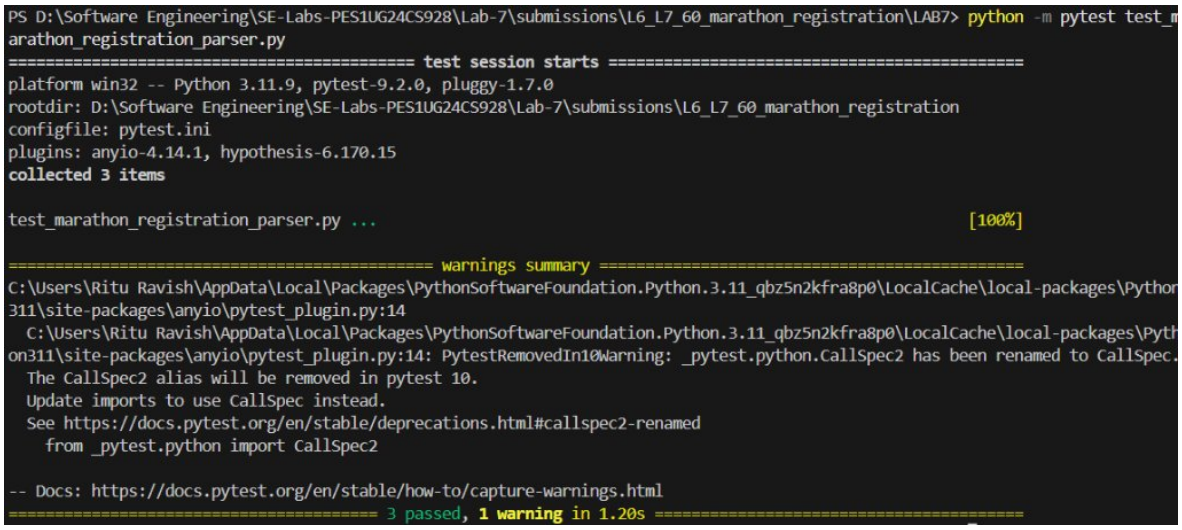
Screenshot 3 (SS3): Running Fixed Application with Valid Inputs

Command: python marathon_registration_parser.py. After applying defensive input validation, normal execution remains intact and outputs match expectations.



Screenshot 4 (SS4): Fuzz Testing Fixed Application with Hypothesis

Command: python -m pytest test_marathon_registration_parser.py. Running the same Hypothesis fuzz tests against the hardened module. All three property-based test cases pass cleanly.



3. Summary of Bugs Found and Fixes Implemented

format_bib_name(name):

- **Bug:** Directly called name.strip().upper()[:12]. When fuzzed with None or non-string types, it crashed with AttributeError: 'NoneType' object has no attribute 'strip'.
- **Fix:** Added isinstance(name, str) validation. If the argument is None or non-string, the function safely returns an empty string "" without crashing.

parse_finish_time(text):

- **Bug:** Unconditionally unpacked text.split(':') into three variables. Failed with AttributeError on None, ValueError on empty strings or strings without two colons, and ValueError on non-integer segments.
- **Fix:** Verified string type, ensured split length is exactly 3, parsed hours/minutes/seconds inside a try-except block, validated ranges (hours ≥ 0, 0 ≤ minutes < 60, 0 ≤ seconds < 60), and returned None on malformed input.

average_split(text):

- **Bug:** Called text.split() and calculated sum(splits) / len(splits). Crashed with AttributeError on None, ZeroDivisionError on empty or whitespace strings, and ValueError on non-numeric tokens.
- **Fix:** Checked string type, ensured tokens exist before division, parsed each token as a positive integer inside a try-except block, and safely returned 0.0 if no valid splits are present.